

Group 1

Mid Term - Exploratory Data Analysis

ITAI-1371

Viswanatha Rao

Team Member Contribution Journal

Viktoriia Kurmisheva:

Handled identifier removal, categorical cleaning, Age and Blood Pressure corrections, and class balancing. Patient_ID and Name were removed because they identify individual patients, while Diagnosis_Code was removed to reduce possible target leakage. Then Age and Blood Pressure values were corrected before later processing. Gender, Diabetic, Smoker, Medications, and Notes were standardized and converted with One-Hot Encoding, with blank medication values represented as 0 and blank Notes values as No_Notes. Random oversampling balanced the training target, followed by final verification and export of the clean dataset. Wrote the group reflection and the What We Will Accomplish document, then uploaded both for the GitHub submission.

Tien Manh Nguyen:

Handled the initial EDA, the 70/30 train-test split, numerical preparation, and City frequency encoding. The split was completed before EDA, and the testing data stayed separate while preprocessing decisions used only the training data. Numerical features were cleaned, missing numerical values were handled with median imputation, and visit dates were converted into year and month features. City labels were standardized and frequency encoded into one City_Frequency column using training-data frequencies rounded to four decimal places. Wrote the Patient Health Records Dataset Proposal and uploaded it along with the main preprocessing Jupyter Notebook for the GitHub submission.

Zaid Alayeddy:

Handled feature engineering, normalization, scaling, and final numerical checks. Pulse Pressure, Mean Arterial Pressure, and the Cholesterol-to-Age Ratio were created from existing health measurements. Skewed numerical distributions were checked and transformed when needed, while Min-Max Scaling placed selected continuous features between 0 and 1 and kept One-Hot Encoded columns binary. Feature ranges, sample outputs, and remaining missing values were verified after processing. Wrote the Team Member Contribution Journal and prepared the separate Before and After Data Processing Jupyter Notebook, then uploaded both for the GitHub submission.